



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-2027

Course Handout

Date: 28/07/2026

In addition to part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course

Course No. : PHA G532
Course Title : QUALITY ASSURANCE & REGULATORY AFFAIRS
Instructor-in-charge : Dr. Gaikwad Anil Bhanudas
Instructor : As mentioned in the time table

Course Description:

Quality control, assurance and management, various parameters for achieving quality pharmaceutical products, application of statistics in quality assurance, reliability, cGMP for pharmaceutical manufacturing, pharmaceutical process validation, drug regulatory affairs, clinical research protocols, new drug applications.

1. Scope and Objective of the course:

This course deals with the basic need of quality in the manufacturing of Pharmaceutical products and its build up with the help of quality control and quality assurance management. Various current Good Manufacturing Practices (cGMP) to be followed in a pharmaceutical organization and its validation procedure are part of the course. This course also covers the regulatory procedures applicable in clinical trials and approval of new drug products.

2. Text Book (T):

- T1: Sidney H. Willig, Murray M. Tuckerman and William S. Hitchings IV, " Good Manufacturing Practices for Pharmaceuticals: A Plan for Total Quality Control" Marcel Dekker, New York, 4th Edn., Vol. 78, 1997.
T2: Bernard T. Loftus and Robert A. Nash, " Pharmaceutical Process Validation" Marcel Dekker, New York, 2nd Edn., Vol. 57, 1993.

Reference Books (R):

- R1: Richard A. Guarino, " New Drug Approval Process" Marcel Dekker, New York, 2nd Edn., Vol. 56, 1993.
R2: Dale H. Besterfield, "Quality Control" Prentice Hall International Inc., New Jersey, 5th Edn., 1998.
R3: Sandy Weinberg, "Good Laboratory Practices" Marcel Dekker, New York, 2nd Edn., Vol. 69, 1995.
R4: Leon Lachman, Herbert L. Lieberman and Joseph L. Kanig " The Theory and Practice of Industrial Pharmacy" Varghese Publ., Bombay, 3rd Edn., 1987.
R5: U.S. Pharmacopeia, U.S. Pharmacopeial Convention Inc., Rockville, MD, 23rd Edn., 1995.



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4. Course Plan:

Lecture #	Module	Topic to be covered	Reference	Learning Objectives
1-8	New Drug Approval Process	a) Target Identification b) Pre-clinical studies c) Investigational New Drug Application d) Phases of clinical trials e) New Drug Application	R1Ch.1-3, 5, 10, 11 Class notes	a) How the new drugs are discovered and developed? Drug Cycle b) Understanding the regulations, requirements, procedures and applications of new drug approval process
8-12	Quality Management	Total Quality Management (TQM)	R2 Ch. 13	a) Different definitions of quality b) Qualitative way of Improving the quality in an industry c) Concepts and implementation of TQM d) Problem Solving Approach e) Characteristics of Leader
12-22	Quality Control	Statistical Quality Control: a) Introduction to quality b) Factors affecting the quality c) Normal Distribution d) Control Chart e) Process capability f) Type of errors g) Six sigma	T1 Ch. 1 R2 Ch. 1 R3 Ch. 1 Class notes	a) Overview of quality control and process in a pharmaceutical industry b) Calculating the central line and control limits c) Identifying the natural and unnatural variation d) Underrating the type of errors in industrial process e) Role of three and sigma in industrial processes
23-27	Preclinical Studies	Good Laboratory Practices:	Class notes	Understanding the significance, implementation





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		a) NGCMA b) GLP certification c) Study Director d) Quality Assurance Unit e) facilities		and guideline of Good Laboratory Practices
28-33	GMP	a) Organization & Personnel b) Buildings & Facilities c) Equipment d) Components, Containers & Closures e) Production & Process control f) Packaging & Labeling control g) Laboratory controls- Reports & Records h) Return Goods & Relabeling	T1 Ch. 3-13	Concepts and tools in the planning, implementation and control of current Good Manufacturing Practices in the Pharmaceutical industry
34-36	Regulations	FDA, ICH guidelines Common technical documents [M4]	Class notes	
37	Pharmaceutical Audit	Quality Audit	T1	a) Definition b) Types of Audit c) Characteristics of a good audit
38-39	ISO	ISO 9000 certification	Class notes	Process of creating standards and significance of ISO 9000 certification
40-42	Validation	Pharmaceutical Process Validation* a) Organization b) Pharmaceutical Products (Solid dosage forms and Sterile products) c) Prospective	T2 Ch. 2-9	Concepts, tools, methods and statistical application in validation of various areas in a Pharmaceutical industry





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		validation d) Retrospective validation & Analysis e) Raw material validation f) Analytical method validation		
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* - Some of the aspects of the above topics will be covered during the laboratory hours.

5. Evaluation Scheme:

No.	Evaluation Component	Duration	Weight-age (%)	Date & Time	Nature of Component
1	Mid Sem	90 min	30		CB
4	Continuous Assessment*		40		OB
5	Comprehensive Exam	120min	30		CB

#Continuous assessment will be based on theory covered in class. Topics and number will be announced in the class. It will be in terms of home assignments, tutorials, projects, laboratory, viva- voce and presentation/ seminars.

Chamber Consultation Hour: To be announced in the class.

Make-up policy: Generally make-up will be considered for regular students only (80% attendance in lecture classes). It is solely dependent on the “genuineness” of the circumstances. The make-up application should be personally given to instructor-in-charge.

Notices: Concerning this course will be displayed on Pharmacy Dept. notice board only.

Instructor-in-Charge
PHA G532



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